

Timeline	FHE Track	IDS Track
Month 1-2	FHE Scheme Selection: • Scheme & Parameter Selection • Library Evaluation & Cross-Compilation Deliverables: • Security specification report • Baseline Performance Benchmarking on Hardware	Threat Matrix & Model Architecture: • Threat specification • Dataset acquisition • Model selection Deliverables: • Threat model document • Baseline architecture design
Month 3-4	Embedded Performance Optimization • Encoding & SIMD Packing • NTT Kernel Acceleration Deliverables: • Optimized Packing & Encoding Module • High-Performance NTT Implementation	Data Engineering & Training • Feature extraction • Baseline training – Centralized Deliverables: • Preprocessing Engine • Model Accuracy Report
Month 5-6	Serialization & V2X Adaptation • Ciphertext Serialization & Communication Protocol	Embedded Porting & Quantization • Model Quantization • On-Device Inference Engine Deliverables: • Quantized Model Weights • Embedded IDS Executable

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Month 7-8	• Aggregation & Functional Logic • Homomorphic Aggregator Deliverables: • Verified Homomorphic Aggregator • Noise Budgeting Tool	Federated Learning & Client Logic • Local Training Loop • Non-IID Handling Deliverables: • FL Client Module • Adaptive Learning Strategy
Month 9-10	System Integration & Stress Testing: • End-to-End Encryption Pipeline • Hardware-in-the-Loop (HiL) Validation Deliverables: • Integrated Encryption/Decryption API for IDS • Performance & Power Consumption Audit Report	Attack Simulation & Red Teaming • Attack Vector Injection • False Positive Tuning Deliverables: • Vulnerability Assessment Report • Final Model Weights (Pre-Encryption Integration)
Month 11-12	Auditing & Project Closure • Privacy Audit & Differential Analysis • Documentation & Final Hardening Deliverables: • Security & Privacy Audit Report • Production-Ready IDS FHE Library	Performance Evaluation & Final Hardening • Accuracy vs. Encryption Overhead • System Hardening & Packaging Deliverables: • Privacy-Utility Trade-off Analysis • Final Project Report & Validated Collaborative IDS System